Signal Processing 91 (2011) 1920–1932

Contents lists available at ScienceDirect

Signal Processing

journal homepage: www.elsevier.com/locate/sigpro

A Bayesian Lasso via reversible-jump MCMC

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ARTICLE INFO

Article history:
Received 17 June 2010
Received in revised form
15 December 2010
Accepted 18 February 2011
Available online 24 February 2011

Keywords:
Sparse signal recovery
Variable selection
Dimensionality reduction
Fully Bayesian modeling
Reversible-jump Markov chain Monte Carlo (RJ-MCMC)
Lasso

ABSTRACT

Variable selection is a topic of great importance in high-dimensional statistical modeling and has a wide range of real-world applications. Many variable selection techniques have been proposed in the context of linear regression, and the Lasso model is probably one of the most popular penalized regression techniques. In this paper, we propose a new, fully hierarchical, Bayesian version of the Lasso model by employing flexible sparsity promoting priors. To obtain the Bayesian Lasso estimate, a reversible-jump MCMC algorithm is developed for joint posterior inference over both discrete and continuous parameter spaces. Simulations demonstrate that the proposed RJ-MCMC-based Bayesian Lasso yields smaller estimation errors and more accurate sparsity pattern detection when compared with state-of-the-art optimization-based Lasso-type methods, a standard Gibbs sampler-based Bayesian Lasso and the Binomial–Gaussian prior model. To demonstrate the applicability and estimation stability of the proposed Bayesian Lasso, we examine a benchmark diabetes data set and real functional Magnetic Resonance Imaging data. As an extension of the proposed RJ-MCMC framework, we also develop an MCMC-based algorithm for the Binomial–Gaussian prior model and illustrate its improved performance over the non-Bayesian estimate via simulations.

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1. Introduction

1.1. Sparse linear models

Consider the following multivariate linear regression model

$$\mathbf{y} = X\boldsymbol{\beta} + \mathbf{e}, \tag{1}$$

where $X_{n\times p}$ represents the design matrix with rows being n measurements and columns being p potential candidate covariates/predictors; $\mathbf{y}_{n\times 1}$ means the response vector of length n; $\boldsymbol{\beta}_{p\times 1}$ means the coefficient vector of length p; and $\mathbf{e}_{n\times 1}$ means the noise vector, following *i.i.d.* Gaussian distribution with zero mean and constant variance σ^2, i.e. $\mathbf{e}_{n\times 1} \sim N(\mathbf{0},\sigma^2 I)$.

Solving the above regression problem statistically involves accurately estimating the coefficients $\boldsymbol{\beta}$ and the noise variance σ^2. *The classical* estimator of $\boldsymbol{\beta}$ in the linear regression context is the well-known Least Squares (LS) estimator. Unfortunately, when dealing with real-world applications which require learning from relatively few data samples when many possible predictors are available, a significant issue, the "curse of dimensionality" (i.e. the difficulty of fitting a model from a few data samples in a high-dimensional feature space), immediately arises with the LS approach. Many biomedical applications suffer from the "curse of dimensionality". One example application is in functional Magnetic Resonance Imaging (fMRI) data. Multivariate autoregressive models (mAR) [15] have been suggested as a way to model brain effective connectivity from neuroimaging data. However, when dealing with fMRI data in practice,

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doi:10.1016/j.sigpro.2011.02.014

traditional full mAR modeling for inferring brain connectivity will break down when the number of nodes (e.g. brain regions-of-interests (ROIs)) is large compared with the number of fMRI time points. For instance, considering a network consisting of 20 ROIs, each increment in the mAR model order will result in an additional 400 parameters to be determined, while only limited observations (typically a few hundred with fMRI experiments) are available.

One approach to overcome the curse of dimensionality in classical regression for model (1) is to attempt regression based on a selection of variables. This requires a procedure of searching among subsets of all possible combinations of explanatory variables to find just a few variables which can adequately explain the dependent variable. In the example of modeling brain networks using fMRI, this variable selection procedure could significantly reduce the number of edges in the learned networks, resulting in sparse connectivity networks. The formulation of variable/model selection problem has been presented concomitantly in the statistics, applied mathematics and signal processing communities, and the procedure may be referred to as sparse linear regression, subset selection in linear regression, or sparse approximation. In this paper, the references are mostly from the statistics literature because here we focus on the Lasso model, which was investigated mainly within the statistics community. Nevertheless, it is worth mentioning that sparse linear approximation of a signal from a set of overcomplete dictionary is also an intensive research topic in both the signal processing and compressed sensing literature [7,34–36] and is sometimes referred to as *basis pursuit* [4]. Here, the *dictionary* is the collection of predictors (i.e. the columns of X). As examples of the application, Peotta et al. [28] represented a sparse coding for the edge part of images; David and Daudet [5] proposed to sparsely generate coefficients of the dictionary consisting of the modified discrete cosine transform (MDCT) atoms of an audio signal.

There have been many variable/model selection methods proposed in the literature from both frequentist and Bayesian perspectives, with a good overview given in [19,30]. With such a wealth of methods, it is difficult to argue which model is universally preferable. Among different methods, Bayesian approaches using Markov chain Monte Carlo (MCMC) have recently become popular [6]. For instance, the "spike and slab" priors on the regression coefficients were proposed in [25]. Using the Laplace or Bernoulli–Gaussian mixture prior on the regression coefficients and introducing latent variables to identify subsets have been the popular choice. Assuming a hierarchical Bayes Gaussian mixture model with latent variables to identify subsets, a Gibbs sampling approach was presented in George and McCullogh [11]. The work in [11] was further explored in [20] by embedding the priors jointly. Several MCMC methods have been compared in [6] for selecting the regression coefficients. In this paper, we are interested in the general category of MCMC-based Bayesian approaches; however, as motivated by the success of Lasso model which is to be discussed shortly, we will employ a Bayesian model different from the above mixture models and propose a fully Bayesian Lasso framework with the RJ-MCMC approach (not the regular MCMC approaches as in the above work).

Among variable/model selection methods, of particular interest are the penalized regression approaches which can unify many techniques with an easy-to-implement framework [9]. The penalized regression can be formulated as optimizing the constrained ℓ_2 loss objective function. Among penalized regression techniques, perhaps the most commonly used one in the statistical learning literature is 'least absolute shrinkage and selection operator', the *Lasso* model [32]. The Lasso estimator $\hat{\boldsymbol{\beta}}_{\text{Lasso}}$ is defined by

$$\hat{\boldsymbol{\beta}}_{\text{Lasso}} = \arg\min \|\mathbf{y} - X\boldsymbol{\beta}\|_2^2 \quad \text{subject to} \quad \|\boldsymbol{\beta}\|_1 \le t, \tag{2}$$

where $\|\boldsymbol{\beta}\|_1 = \sum_{j=1}^{p} |\beta_j|$ and $\|\boldsymbol{\beta}\|_2 = \sqrt{\sum_{j=1}^{p} \beta_j^2}$, the ℓ_1 and ℓ_2 norms of $\boldsymbol{\beta}$, respectively. Here $t \ge 0$ is a shrinkage tuning parameter. One advantage of the Lasso is that it can shrink some coefficients and meanwhile set others to 0. In (2) since both the objective function and the constraint are convex functions, the Lasso estimate can be solved by standard convex optimization techniques [2]. Efron et al. [8] recently proposed a computationally efficient algorithm for obtaining both the Least Angle Regression (Lar) and the Lasso estimates.

The penalized approach in (2) has an alternative Bayesian interpretation. As noted in Tibshirani [32], Lasso estimates can be interpreted as *Maximum A Posteriori* (MAP) estimates with the regression coefficients possessing independent Laplace (a.k.a. double-exponential) priors. More recently, Hunter and Li [18] proposed a more general optimization approach, the Minorize-Maximize (MM) algorithm [21], to transfer the problem of maximizing the posterior function w.r.t. the Laplace prior to sequentially maximizing its quadratic surrogate functions. Motivated by this connection between Lasso estimates and the Bayesian interpretation for the Laplace prior, several Laplace-like priors have been recently proposed for promoting sparsity, e.g. a mixture of delta-mass at 0 and the Laplace prior was studied in [37] and Jeffrey's non-informative mixing distribution on the prior of $\boldsymbol{\beta}$ in [10]. The popularity and the good performance of the Lasso model motivates us to employ a Laplace prior for the regression coefficients in our proposed RJ-MCMC-based Bayesian approach, a Bayesian Lasso estimator. The observations in [27] actually suggested potential advantages of the Laplace prior over a Gaussian (or a Student-t) prior.

1.2. Related work and our contributions

It must be emphasized that the non-Bayesian Lasso and Lasso-like approaches have one aspect in common: they are optimization methods with the goal of determining the model parameters that maximize some objective function. Meanwhile, the number of variables set to be zero in these methods critically depends on the tuning shrinkage parameter, where its value can generally be selected to minimize the generalized cross-validation

errors. In this paper, apart from those Lasso-like optimization methods, we propose a new fully Bayesian framework to deal with the Lasso objective function. Such a fully Bayesian approach with the Laplace prior on $\boldsymbol{\beta}$, referred as *a Bayesian Lasso*, does not require cross-validation (CV) type methods to determine the optimum shrinkage parameter as in Lasso, since a non-informative prior is given for the shrinkage controlling parameter and its posterior distribution is completely derived from the observed data. By integrating parameters w.r.t. their posterior distributions, the proposed Bayesian Lasso estimator has a different posterior distribution from the ordinary Lasso (Laplace prior), and can yield more robust estimates.

Very recently, work has been proposed in the direction of Bayesian Lasso [27]. In [27], with a conditional Gaussian prior on $\boldsymbol{\beta}$ and the non-informative scale-invariant prior on the noise variance being assumed, a Bayesian Lasso model is proposed and a simple Gibbs sampler is implemented. It is shown that the Bayesian Lasso estimates in [27] are strikingly similar to those from the ordinary Lasso. Since this Bayesian Lasso in [27] involves the inversion of the covariance matrix of a block coefficients at each iteration, the computationally complexity prevents its practical application with, say, hundreds of variables. Moreover, similar to the regular Lasso, the Bayesian Lasso in [27] uses only one shrinkage parameter t to both control model size and shrink estimates. Nonetheless, it is arguable whether the two simultaneous effects can be well handled by a single tuning parameter [24]. To mitigate this non-separability problem, Mutshinda and Sillanpää [26] proposed an extended Bayesian Lasso model by assigning a more flexible, covariate-adaptive penalization on top of the Bayesian Lasso in the context of Quantitative Trait Loci (QTL) mapping. Alternatively, introducing different sources of sparsity promoting priors on both coefficients and their indicator variables have been studied, e.g. in [29], where a normal-Jeffreys scaled-mixture prior on coefficients and an independent Bernoulli prior with small success probability on the binary index vector are combined. Motivated by this observation, we introduce two parameters in the proposed Bayesian Lasso model to separately control the model selection and estimation shrinkage issues in the spirit of [24,37], and propose a Poisson prior on the model size together with the Laplace prior on $\boldsymbol{\beta}$ to identify the sparsity pattern. Since the proposed joint posterior distribution is highly non-standard and a standard MCMC is not applicable, we employ a reversible-jump MCMC (RJ-MCMC) to obtain the proposed Bayesian Lasso estimates by simultaneously performing model averaging and parameter estimation. It is worth emphasizing that, though RJ-MCMC algorithms have been developed in the literature before model selection and estimation purposes (e.g. Andrieu and Doucet [1] proposed a hierarchical Bayesian model and developed an RJ-MCMC algorithm for joint Bayesian model selection and estimation of noisy sinusoids; similarly Sillanpää and Arjas [31] proposed an accelerated truncated Poisson process model for Bayesian QTL mapping), these methods are not intended for promoting sparse models, whereas our model utilizes sparsity promoting priors in conjunction with the discrete prior on the model size.

As we show later, the proposed fully Bayesian Lasso framework provides estimation performance improvements when compared with Lasso, the Gibbs sampler-based Bayesian Lasso in [27] and the Binomial–Gaussian model in [12]. When handling the nearly singular case ($p \approx n$), the performance improvement from the proposed Bayesian Lasso estimate is even more significant. As a side benefit, we also extend the proposed RJ-MCMC estimation framework to the Binomial–Gaussian model in [12], and the developed BG-MCMC approach yields significant performance improvements over the original non-Bayesian approach in [12].

1.3. Notations

We now define some notations to be used throughout the paper. Let $\boldsymbol{\gamma}$ be a p-length binary vector where ones denote non-zero coefficients and zeros denote zero coefficients. Equivalently, position of ones in $\boldsymbol{\gamma}$ can be thought as the *active set* or *support* of a linear regression model while position of zeros is called the *inactive set*. $|\boldsymbol{\gamma}|$ is used to denote the number of non-zeros in $\boldsymbol{\gamma}$, meaning the cardinality of the support of $\boldsymbol{\gamma}$. Some special functions and probability density functions are listed in Table 1.

The rest of the paper is organized as follows. In Section 2, we first describe the new hierarchical, fully Bayesian Lasso model, and then propose an RJ-MCMC algorithm to simulate this posterior distribution for computing the unbiased minimum variance estimator of the regression coefficient vector. Simulations are carried out in Section 4 to evaluate the performance of the proposed approach. Section 5 presents the results on a diabetes data set. In Section 6, we apply the proposed method to real fMRI data to demonstrate the applicability of the proposed Bayesian Lasso method.

2. A fully Bayesian Lasso model

2.1. Prior specification

The proposed fully Bayesian model has the basic structure of a standard linear regression model in (1), with the addition of priors over the parameters to be estimated. Objective hyper-priors are chosen on the sparsity promoting priors.

Firstly, to achieve the parsimonious estimation goal, we assign sparsity promoting priors on each of the non-zero coefficients β_j' s. Here, independent Laplace priors

Table 1
Special functions and probability density functions used in the paper.

Name	Functional form
Gamma function	$\Gamma(a)=\int_0^\infty t^{a-1}\exp(-t)\,dt$
Beta function	$B(a,b)=\Gamma(a)\Gamma(b)/\Gamma(a+b)$
Gamma distribution	$\mathrm{Ga}(x;a,b)=(b^a/\Gamma(a))x^{a-1}\exp(-bx)$
Inverse Gamma distribution	$\mathrm{IG}(x;a,b)=(b^a/\Gamma(a))x^{-(a+1)}\exp(-b/x)$
Beta distribution	$\mathrm{Beta}(x;a,b)=x^{a-1}(1-x)^{b-1}/B(a,b)$

are assumed for $\boldsymbol{\beta}$, i.e. each component β_j in the active set $\boldsymbol{\gamma}$ with $|\boldsymbol{\gamma}|=k$ follows the distribution

$$\pi(\beta_j|\tau,\boldsymbol{\gamma})=\frac{1}{2\tau}\exp\left(-\frac{|\beta_j|}{\tau}\right), \tag{3}$$

where τ is a shrinkage tuning parameter for β_j with $j\in\boldsymbol{\gamma}$. Otherwise, $\beta_j\equiv 0$ for $j\notin\boldsymbol{\gamma}$.

Since in practice we may not have prior knowledge regarding how much shrinkage amount should be put on the coefficients before performing any experiment, it is reasonable to also assign a non-informative prior on the hyper-parameter τ. A non-informative prior for the noise variance σ^2 is given based on the same rationale. Together, traditional non-informative priors are put on the higher level as

$$p(\tau,\sigma^2)\propto(\tau\sigma)^{-1}. \tag{4}$$

Due to Lindley's paradox [22], one needs to be very careful to assign improper priors for τ and σ^2 when performing model selection/averaging. Since we do not allow the null model with no predictor, τ and σ^2 are the common parameters for all possible sub-models. Hence, the under-determined proportional constants are the same for all sub-models and, therefore, do not affect the model comparisons based on the posteriors.

Further, prior probabilities are also assigned to each possible sub-model. Specifically, a sub-model containing k predictors follows a right-truncated Poisson distribution at p,

$$p(k|\lambda)=\frac{e^{-\lambda}\lambda^k}{Ck!},$$

where C is a normalization constant, and $k=1,\ldots,p$. Within the set of sub-models having k predictors, each sub-model is assumed to be with equal prior probability. To complete the prior specification, the parameter λ is also assumed to follow a non-informative prior as

$$p(\lambda)\propto\lambda^{-1}.$$

Therefore, the posterior distribution of the parameter vector $(\boldsymbol{\beta},\tau,\sigma^2,\boldsymbol{\gamma},\lambda)$ can be expressed as (up to a multiplicative constant)

$$p(\boldsymbol{\beta},\tau,\sigma^2,\boldsymbol{\gamma},\lambda|\mathbf{y},X)\propto\frac{e^{-\lambda}\lambda^{k-1}}{\binom{p}{k}k!}\tau^{-(k+1)}\sigma^{-(n+1)}$$
$$\exp\left(-\frac{\|\boldsymbol{\beta}\|_1}{\tau}-\frac{\|\mathbf{y}-X\boldsymbol{\beta}\|_2^2}{2\sigma^2}\right). \tag{5}$$

The derivation detail of (5) can be found in the Appendix A. It is further noted that, to reduce the computational cost, the parameters τ,σ^2,and,λ can be analytically integrated out. More specifically, by noting the normalization constants of the respective Gamma and inverse Gamma p.d.f.'s (Table 1), we can show that

$$\pi(\boldsymbol{\beta},\boldsymbol{\gamma}|\mathbf{y},X)\propto\Gamma(k)B(k,p-k+1)\|\boldsymbol{\beta}\|_1^{-k}\|\mathbf{y}-X\boldsymbol{\beta}\|_2^{-(n-1)}. \tag{6}$$

Comparing (6) with the posterior distribution of the standard Lasso reveals interesting observations. Eq. (6) comprises two parts: $\|\boldsymbol{\beta}\|_1^{-k}$ is the prior information and $\|\mathbf{y}-X\boldsymbol{\beta}\|_2^{-(n-1)}$ is the likelihood. These two parts are linked through polynomial terms, representing a different weighting scheme from Lasso. By integrating out the nuisance parameters analytically, a more stable estimator is possible, in addition to the advantage of requiring fewer computations.

It is obvious that this joint posterior distribution is in a non-standard form and there is no closed-form analytic expression of $\mathbb{E}(\boldsymbol{\beta}|\mathbf{y},\mathbf{X})$ w.r.t. its posterior distribution. Hence, we must resort to simulation-based approaches to compute the numeric estimates. In this paper, we will develop a Markov chain Monte Carlo (MCMC) type of estimation method.

Remark 1. For the standard Lasso approach, there is one shrinkage parameter t in (2) to both control model size and shrink estimates, but for some applications, it is desirable to separate those two effects. Here, in the proposed framework, we have two parameters (i.e. τ and λ) to separately control the model selection and estimation shrinkage issues. Roughly speaking, a Poisson prior on the size of active set k controls the size of the expected number of selected predictors and a Laplace prior on $\boldsymbol{\beta}$ recovers the non-zero coefficients which can best represent the full model conditioned on k.

3. Bayesian computation

Algorithm 1. The proposed RJ-MCMC based Bayesian Lasso

Input: The number of iterations T. Random walk step size ε.
Data: X and $\mathbf{y}$.
Output: $\{\boldsymbol{\theta}^{(t)}=(\boldsymbol{\beta}^{(t)},\boldsymbol{\gamma}^{(t)})|t\in\{0,\ldots,T\}\}$.

begin
  Initialization : set $\boldsymbol{\theta}^{(0)}=(\boldsymbol{\beta}^{(0)},\boldsymbol{\gamma}^{(0)})$ and $t=1$.
  repeat
    if $k^{(t-1)}=1$ **then**
      $k^{(t)}\leftarrow k^{(t-1)}+U(\{0,1\})$.
    else if $k^{(t-1)}=p$ **then**
      $k^{(t)}\leftarrow k^{(t-1)}-U(\{0,1\})$.
    else
      $k^{(t)}\leftarrow k^{(t-1)}+U(\{-1,0,1\})$.
    end
    Sample $s\sim N(0,\varepsilon^2)$.
    $K\leftarrow\boldsymbol{\gamma}^{(t-1)}$ and $K^c\leftarrow\{1,\ldots,p\}\backslash K$.
    if $k^{(t-1)}=k^{(t)}$ **then**
      Sample $j\sim U(K)$.
      Update $\beta_j^{(t)}\leftarrow\beta_j^{(t-1)}+s$ with an MH step, details in Section 3.2.
    else if $k^{(t)}=k^{(t-1)}+1$ **then**
      Sample $j\sim U(K^c)$.
      Perform a " birth" move and update $\beta_j^{(t-1)}$, details in Section 3.3.
    else
      Sample $j\sim U(K)$.
      Perform a " death" move and update $\beta_j^{(t-1)}$, details in Section 3.3.
    end
    $t\leftarrow t+1$.
  until $t=T$.
end

Since the joint posterior distribution in (6) contains both discrete and continuous parameters, a closed-form solution of the unbiased minimum variance estimator is infeasible: the posterior expectation of coefficients $\mathbb{E}(\boldsymbol{\beta}|X,\mathbf{y})$ and the posterior probability of the inclusion of

coefficients $\mathbb{E}(\gamma|X,\mathbf{y})$. Moreover, standard MCMC is not applicable in this case since the model dimension is not fixed. To address this difficulty, we propose a hybridized MCMC sampler to simultaneously perform model averaging and parameter estimation. Our proposed algorithm falls into the RJ-MCMC umbrella [14]. RJ-MCMC is a powerful prototype that creates MCMC algorithms for variable dimensional models and may be better than separate within-model MCMC runs if we aim at making joint inference about the models and their parameters. Moreover, running separate MCMC for each model is computationally prohibitive for large scale problems. The proposed algorithm is summarized in Algorithm 1, with more details given in the following paragraphs.

3.1. Design of model transition

It is clear that the distribution of $\boldsymbol{\beta}$ depends on the model dimensionality. For example, deleting predictors will force their corresponding index set to zeros. Here, we propose three types of model moves as following:

1. $\gamma\to\gamma$; 2. $\gamma\to\gamma'$; 3. $\gamma'\to\gamma$.

To smoothly move between models and allow fast mixing, we design local jumps. At each sampling step, the current model is only allowed to stay in the same dimension or move to its neighboring models. As illustrated in Fig. 1, the proposed model has either the same dimension as the previous one, or one predictor added or deleted from the current model. Moreover, we assign each possibility with equal probabilities, as

$$p(\gamma\to\gamma)=\tfrac{1}{3},$$

$$p(\gamma\to\gamma')=\frac{1}{3(p-k)},$$

$$p(\gamma'\to\gamma)=\frac{1}{3(k+1)},$$

for $k=2,\ldots,p-2$ and $|\gamma|=k$ and $|\gamma'|=k+1$. The boundary models need slightly different probabilities. For $|\gamma|=1$, we do not allow the null model with no predictor at all. Hence for a model with just one predictor, it can only stay in one-dimensional or move to two-dimensional, each with probability $\frac{1}{2}$. Namely, for $|\gamma|=1$ and $|\gamma'|=2$

$$p(\gamma\to\gamma)=\tfrac{1}{2},$$

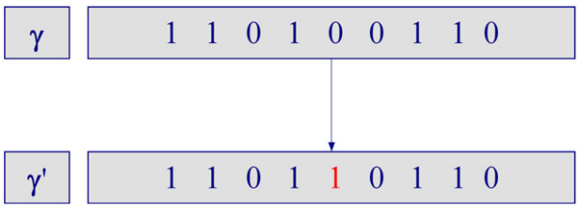


Fig. 1. An illustration of model jumping from $\gamma\to\gamma'$ with $|\gamma|=5$ and $|\gamma'|=6$. New predictors (in red) are created from current model. A position with 1 indicates a non-zero coefficient, 0 denotes current model excludes this coefficient. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

$$p(\gamma\to\gamma')=\frac{1}{2(p-1)}.$$

Similarly, for $|\gamma|=p$ and $|\gamma'|=p-1$, we have

$$p(\gamma\to\gamma)=\tfrac{1}{2},$$

$$p(\gamma\to\gamma')=\frac{1}{2p}.$$

3.2. A usual Metropolis–Hastings update for unchanged model dimension

For the models with unchanged dimension, the standard Metropolis–Hastings (MH) algorithm is used to update $\boldsymbol{\beta}$ [17]. Specifically at iteration t, a predictor at position $j\in\gamma$ to be updated is randomly selected from current non-zero coefficients. Then, a proposal distribution $q(\boldsymbol{\theta}^{(t)},\boldsymbol{\theta}')$ is chosen to update this predictor, where $\boldsymbol{\theta}^{(t)}$ is current parameter estimate and $\boldsymbol{\theta}'$ is the proposed parameter. Here, a Gaussian random walk (RW) is used as our proposal, $N(0,\varepsilon^2)$, with some fixed small step size ε. Set $\boldsymbol{\beta}'=\boldsymbol{\beta}+u\mathbf{e}_j$ where $\mathbf{e}_j$ is the jth standard Euclidean basis. Then the acceptance probability becomes

$$\min\left\{\left(\frac{\|\boldsymbol{\beta}'\|_1}{\|\boldsymbol{\beta}\|_1}\right)^{-k}\times\left(\frac{\|\mathbf{y}-X\boldsymbol{\beta}'\|_2}{\|\mathbf{y}-X\boldsymbol{\beta}\|_2}\right)^{-(n-1)},1\right\}. \tag{7}$$

3.3. A birth-and-death strategy for changed model dimension

Since there is no concept of metric structure and compactness as in the Euclidean space for trans-dimensional jumps in $\boldsymbol{\Theta}$, designing an optimal or even a valid proposal is not an easy task. The standard MCMC optimal *scaling* proposal has no analog for reversible jump moves [3]. For the model jumping moves, there are two commonly used proposals: (i) *birth-and-death* and (ii) *split-and-merge*. The birth-and-death is a simple form of model transformation: In the birth step, a new predictor is added to the current model, by generating parameters of a new predictor from a prior distribution; in the death step, a predictor is removed from current model, and the reversibility constraint must be satisfied according to the *detailed balance* equation. Green [14] showed that if there exists a σ-finite symmetric measure μ, with respect to which $\pi(d\boldsymbol{\theta})q(\boldsymbol{\theta},d\boldsymbol{\theta}')$ is absolutely continuous with π is our target posterior distribution in (6), then the detailed balance condition holds for all Borel subsets $B,B'\subset\mathcal{B}(\boldsymbol{\Theta})$,

$$\int_{B\times B'}\alpha(\boldsymbol{\theta},\boldsymbol{\theta}')\rho(\boldsymbol{\theta},\boldsymbol{\theta}')\mu(d\boldsymbol{\theta},d\boldsymbol{\theta}')=\int_{B'\times B}\alpha(\boldsymbol{\theta}',\boldsymbol{\theta})\rho(\boldsymbol{\theta}',\boldsymbol{\theta})\mu(d\boldsymbol{\theta}',d\boldsymbol{\theta}), \tag{8}$$

if the acceptance ratio $\alpha(\boldsymbol{\theta},\boldsymbol{\theta}')$ is chosen to be

$$\alpha(\boldsymbol{\theta},\boldsymbol{\theta}')=\frac{\rho(\boldsymbol{\theta}',\boldsymbol{\theta})}{\rho(\boldsymbol{\theta},\boldsymbol{\theta}')}\wedge 1, \tag{9}$$

where the extended μ-integrable function ρ is the Radon–Nikodym derivative of $\pi\times q$ with respect to μ. It is easy to see that the unchanged model dimension update $\gamma\to\gamma$ is

just a special case by taking μ as the Lebesgue measure on $\mathcal{B}(\mathbb{R}^k)\otimes\mathcal{B}(\mathbb{R}^k)$, the product Borel σ-algebra on $\mathbb{R}^k$.

For a birth move, a new predictor is created by random generation from the inactive set. The proposed predictors are accepted with the probability given by the *generalized Metropolis–Hasting ratio*. More specifically, a coefficient is randomly generated outside the current support set (i.e. generating a $j\in\boldsymbol{\gamma}^c$ and setting the value of this coefficient with a zero-mean Gaussian realization u). Since the model dimension is augmented by generating an additional variable u, there is a Jacobian term for the acceptance probability of the birth move which is 1 in this case.

By putting the posterior ratio computed from (6) and model transition probabilities together into (9), the acceptance probability is given by

$$\min\left\{\frac{k^2}{p-k}\times\frac{\|\boldsymbol{\beta}'\|_1^{-(k+1)}\|\mathbf{y}-X\boldsymbol{\beta}'\|_2^{-(n-1)}}{\|\boldsymbol{\beta}\|_1^{-k}\|\mathbf{y}-X\boldsymbol{\beta}\|_2^{-(n-1)}}\times\frac{p(\boldsymbol{\gamma}'\to\boldsymbol{\gamma})}{p(\boldsymbol{\gamma}\to\boldsymbol{\gamma}')},\times N(u;0,\varepsilon^2)^{-1},1\right\}, \tag{10}$$

where $N(u;\mu,\varepsilon^2)$ means the Gaussian density $N(\mu,\varepsilon^2)$ evaluated at u.

Similarly, the death move is simply the reverse of the birth move. The acceptance probability for $\boldsymbol{\gamma}\to\boldsymbol{\gamma}'$ with $|\boldsymbol{\gamma}|=k$ and $|\boldsymbol{\gamma}'|=k-1$ is given by

$$\min\left\{\frac{p-(k-1)}{(k-1)^2}\times\frac{\|\boldsymbol{\beta}'\|_1^{-(k-1)}\|\mathbf{y}-X\boldsymbol{\beta}'\|_2^{-(n-1)}}{\|\boldsymbol{\beta}\|_1^{-k}\|\mathbf{y}-X\boldsymbol{\beta}\|_2^{-(n-1)}}\times\frac{p(\boldsymbol{\gamma}'\to\boldsymbol{\gamma})}{p(\boldsymbol{\gamma}\to\boldsymbol{\gamma}')},\times N(u;0,\varepsilon^2),1\right\}. \tag{11}$$

Eqs. (10) and (11) ensure the reversibility of the constructed Markov chain by (8).

Remark 2. If one is also interested in the nuisance parameters τ, σ^2 and λ, it is easy to extend the current algorithm to include embedded Gibbs samplers. Since the full conditional distributions of τ, σ^2, and λ can be given in closed forms, the Gibbs sampler [13] is used to simulate these parameters from their posterior distributions,

$$\tau|\cdot\sim\mathrm{IG}\left(k,\sum_{j\in\boldsymbol{\gamma}}|\beta_j|\right), \tag{12}$$

$$\sigma^2|\cdot\sim\mathrm{IG}\left(\frac{n-1}{2},\frac{\|\mathbf{y}-X\boldsymbol{\beta}\|_2^2}{2}\right), \tag{13}$$

$$\lambda|\cdot\sim\mathrm{Ga}(k,1). \tag{14}$$

Here $|\cdot$ means the conditional distribution given everything else, including data and other parameters.

4. Simulations

4.1. Setup

Data sets of 100 data points are simulated, with a series of different model dimensions. The value of p is set to be 15, 45, and 90. Each setting has a specific sparse structure of the coefficients, and we denote a particular setting by a format such as 30/90, meaning that 30 out of the total 90 predictor coefficients are non-zeros. For instance, for the 90 dimensional case, we set the coefficients to be

$$(\underbrace{3,\ldots,3}_{10\text{ times}},\underbrace{0,\ldots,0}_{20\text{ times}},\underbrace{1.5,\ldots,1.5}_{10\text{ times}},\underbrace{0,\ldots,0}_{20\text{ times}},\underbrace{2,\ldots,2}_{10\text{ times}},\underbrace{0,\ldots,0}_{20\text{ times}}).$$

Although independent sparsity promoting priors are used (3), correlations between predictors are also introduced and compared with uncorrelated data and their influence on the performance of various algorithms is explored. We set the correlation level to be 0.5 in our simulations. Several standard linear model selection approaches are compared, including the Lasso [32] (and its variant *Gauss-Lasso* for extending Lasso to accommodate both model selection and regression objectives, where Lasso is used as model selection followed by a least squares estimate based on the selected model), Lar [8] (and its variant *Gauss-Lar*, where Lar is used as model selection followed by a least squares estimate based on the selected model), a Gibbs sampler-based Bayesian Lasso [27] and the Binomial–Gaussian (BG) model [12]. To examine the effects on the overall performance of the proposed Poisson–Laplace model and the proposed MCMC estimation approach, we also extend the BG model in [12] to the fully Bayesian framework, and develop a corresponding MCMC algorithm, referred as *BG-MCMC*, with details given in Appendix B. We shall see that the proposed BG-MCMC method provides significant performance improvements over the original non-Bayesian method in [12]. The shrinkage tuning parameter t for the Lasso and Lar is determined by 10-fold CV with the minimal prediction errors. The proposed BG-MCMC and the Bayesian Lasso estimators are initialized at the LS estimate of the full model and run for 100,000 iterations with the first half runs being discarded as warm-up. The step size ε of RW is 0.05. The performances of the coefficient estimates are measured by the Root Mean Squared Errors (RMSEs) and all results are averaged over 100 simulations.

4.2. Empirical performance comparisons

The RMSE performances of the eight models are summarized in Table 2. Several observations can be summarized as follows:

First, performances of the Lasso and Lar are similar to each other; this is in particular pronounced in lower dimension cases. This is because the Lasso (implemented in a modified Lar algorithm [8]) rarely drops variables from the active set and hence is very similar to the Lar. Further, compared with the Lasso and Lar, the proposed Bayesian Lasso with RJ-MCMC algorithm consistently yields much smaller RMSE and smaller estimation variability. This observation is even more significant for the $p\approx n$ cases where the MLE can be highly unstable. The reduced estimation variability is likely due to the fact that the parameters are estimated based on averaged models, rather than conditioning on a single *best* plausible model given by the penalized MLE principle. Moreover, since the Lasso uses one tuning parameter (t in (2)) to simultaneously select variables and shrink estimates, the estimation may be shrunk along with the decreased model size. However, for our proposed approach, since two parameters (τ and λ) incorporate together to control these

Table 2
RMSEs averaged over 100 simulations for $n=100$. The number in the bracket is the standard deviations of the estimated RMSEs. Methods under comparison are the Lasso [32], Gauss-Lasso, Lar [8], Gauss-Lar, Gibbs sampler based Bayesian Lasso [27], Binomial–Gaussian (BG) [12], proposed BG-MCMC and proposed Bayesian Lasso (BLasso) with RJ-MCMC algorithm.

	No correlation			**Correlation=0.5**		
	3/15	15/45	30/90	3/15	15/45	30/90
Lasso [32]	0.728	1.023	7.968	0.561	0.859	9.895
	(0.146)	(0.158)	(2.441)	(0.116)	(0.163)	(2.360)
Gauss-Lasso	0.153	0.559	5.751	0.280	0.765	9.848
	(0.070)	(0.125)	(3.059)	(0.137)	(0.145)	(5.073)
Lar [8]	0.728	1.021	6.250	0.560	0.868	8.251
	(0.146)	(0.159)	(2.349)	(0.116)	(0.172)	(2.668)
Gauss-Lar	0.153	0.561	3.602	0.280	0.815	6.732
	(0.070)	(0.123)	(2.700)	(0.137)	(0.155)	(3.503)
Gibbs sampler [27]	0.376	0.832	1.837	0.499	1.126	2.422
	(0.087)	(0.121)	(0.294)	(0.112)	(0.170)	(0.375)
BG [12]	0.165	0.429	10.362	0.199	0.604	11.537
	(0.077)	(0.103)	(3.254)	(0.103)	(0.130)	(4.437)
BG-MCMC	0.180	0.435	0.704	0.220	0.577	0.909
	(0.072)	(0.106)	(0.129)	(0.107)	(0.147)	(0.149)
Proposed BLasso	0.157	0.417	0.708	0.199	0.577	1.497
	(0.068)	(0.080)	(0.235)	(0.092)	(0.111)	(0.959)

Table 3
F-scores of estimated sparsity patterns averaged with standard deviations in brackets over 100 simulations for $n=100$. Methods under comparison are the Lasso [32], Lar [8], Gibbs sampler based Bayesian Lasso [27], Binomial–Gaussian (BG) [12], the proposed BG-MCMC and the proposed Bayesian Lasso (BLasso) with RJ-MCMC algorithm.

	No correlation			**Correlation=0.5**		
	3/15	15/45	30/90	3/15	15/45	30/90
Lasso [32]	0.996	0.888	0.713	0.927	0.802	0.599
	(0.025)	(0.062)	(0.179)	(0.099)	(0.049)	(0.133)
Lar [8]	0.996	0.887	0.798	0.927	0.763	0.673
	(0.025)	(0.063)	(0.119)	(0.099)	(0.046)	(0.099)
Gibbs sampler [27]	0.964	0.612	0.528	0.929	0.572	0.517
	(0.070)	(0.066)	(0.037)	(0.109)	(0.062)	(0.041)
BG [12]	0.850	0.505	0.230	0.791	0.393	0.141
	(0.097)	(0.066)	(0.017)	(0.048)	(0.086)	(0.045)
BG-MCMC	0.993	0.993	0.991	0.994	0.998	0.999
	(0.031)	(0.014)	(0.012)	(0.028)	(0.008)	(0.004)
Proposed BLasso	1.000	1.000	1.000	1.000	1.000	0.979
	(0.000)	(0.003)	(0.004)	(0.000)	(0.000)	(0.070)

two effects, it is more flexible and likely to obtain an unbiased estimate. To support this claim, we reported the RMSE results of Gauss-Lasso and Gauss-Lar in Table 2 and also reported the estimated sparsity patterns in Lasso and Lar in Table 3. These two tables together show that the major source of RMSEs of Lasso and Lar comes from the errors made in the model selection stage.

The proposed RJ-MCMC-based Bayesian Lasso also consistently yields better estimation accuracy than the Gibbs sampler-based Bayesian Lasso. The Gibbs sampler method yields the MSEs between the proposed RJ-MCMC method and Lasso/Lar.

The BG method in [12] has comparable, slightly worse performance over the proposed Bayesian Lasso in lower dimension regimes. However, when the model size increases comparable to the data size (e.g. $p=90$ and $n=100$), the performance of the BG method substantially degrades. The major advantage of the BG model is that the marginal likelihood of the model can be given in closed-form, conditioned on the known active set for a model. However, since comparing marginal likelihood of all possible models requires the enumeration of the 2^p possible models, an exhaustive search is not computationally feasible so a common means to approximate the exact solution is to adopt a stepwise searching strategy. As used in [12], forward selection is used to traverse between models, however, false predictors introduced in earlier stages of the algorithm cannot be eliminated at a later stage. Degraded performance of BG is especially pronounced when predictors are highly correlated or the sample size is not large enough. Hence, for medium/large scale problems with no special structure (e.g. orthogonality among predictors), the BG method is not a good choice for model selection in practice. In contrast, the stochastic search algorithm proposed in this paper successfully avoids being trapped by sub-optima with the price of increased computational cost. Empirically, we observe that the proposed fully Bayesian algorithms can accurately estimate the model and associated parameters with a reasonable sampling size.

By extending the proposed fully Bayesian framework to the BG model [12], we further confirm the strength of the MCMC approach. We derive an MCMC algorithm for the BG model, as in Appendix B. Many parameters can be integrated out because of the Gaussianality. The main interesting quantity is the model index parameter, the posterior of which can be viewed as the posterior probability of including the corresponding coefficient. It is seen from Table 2 that the derived BG-MCMC achieves similar performances as the proposed RJ-MCMC-based Bayesian Lasso.

We note empirically that there is not much room for improvement for the two fully Bayesian approaches, the proposed BLasso and BG-MCMC. With the assumption that we are given an oracle revealing the location of the true non-zero coefficients, it is easy to see that the optimal least squares estimator has an RMSE converging to $\sqrt{(k/n)}\sigma$ for large n. In our setup, with $\sigma^2=1$, the fundamental information-theoretic limits of RMSEs in Table 2 are 0.173, 0.387, 0.548, respectively. In our simulations, we empirically observe that the performances of the proposed RJ-MCMC-based Bayesian Lasso and BG-MCMC approximate the lower estimation error bound. Therefore, we suggest that there is a substantial advantage over greedy search and optimization-based methods by using the proposed fully Bayesian framework coupled with the stochastic search.

We also examine the sparsity patterns recovered by different algorithms under consideration. For BG, Lasso and Lar, sparsity patterns are characterized by the locations of non-zero coefficients. For BG-MCMC and the proposed Bayesian Lasso with RJ-MCMC, a non-zero coefficient is declared when its posterior probability

passes the threshold 0.5. For the Bayesian Lasso based on the Gibbs sampler, since the distribution of $\boldsymbol{\beta}$ is assumed to have a conditional zero-mean normal distribution, we calculate its two-side tail probabilities that exceed the posterior estimate $|\hat{\boldsymbol{\beta}}_j|$ and then compare the tail probabilities with a significance level 0.1. To compare the algorithms, we evaluate the performances of support recovery in terms of their *F*-scores which combine the precision and recall (true positive rate) measures. More specifically, the precision *P* is the fraction of detected true positive among all identified positives, whereas the recall *R* is the identified true positive ratio to the total number of true positives. The *F*-score is defined as the harmonic mean of the precision and recall, i.e.

$$F=\frac{2PR}{P+R}. \tag{15}$$

The closer *F*-score of an algorithm is to 1, the better performance it has.

It is clear from Table 3 that the proposed BLasso and BG-MCMC yield the best and stablest performances, with the proposed BLasso slightly outperforms BG-MCMC. Lasso, Lar, Gibbs sampler, BG have comparable support detecting capability in lower dimensional problems; however, their performances significantly degrade as the problem size gets larger.

4.3. Convergence analysis

We now prove that the proposed RJ-MCMC framework in Algorithm 1 converges to the posterior distribution of $(\boldsymbol{\gamma},\boldsymbol{\beta})$ given in (6). Let $M=\{(\boldsymbol{\gamma}^{(i)},\boldsymbol{\beta}^{(i)})\}_{i\in\mathbb{N}}$ be the Markov chain constructed by Algorithm 1 such that the detailed balance condition (8) implies $\pi(\boldsymbol{\gamma},\boldsymbol{\beta}|\mathbf{y})$ is an invariant distribution for *M*. With this target distribution, it suffices to show that *M* is ergodic with respect to $\pi(\boldsymbol{\gamma},\boldsymbol{\beta}|\mathbf{y})$ [33]. This is equivalent to show that *M* is $\pi(\boldsymbol{\gamma},\boldsymbol{\beta}|\mathbf{y})$-irreducible and aperiodic. Aperiodicity is obvious and the only part we need to argue is the $\pi(\boldsymbol{\gamma},\boldsymbol{\beta}|\mathbf{y})$-irreducibility. The idea of showing the irreducibility of *M* is to find a particular path that monotonically shrinks the model size to one with certain positive probability. We can assume without loss of generality that the only predictor in the destination model is the first one. Let $K(\boldsymbol{\gamma},\boldsymbol{\beta};\boldsymbol{\gamma}',d\boldsymbol{\beta}')$ be the transition kernel of *M*:

$$P(\boldsymbol{\gamma}',\boldsymbol{\beta}'\in B|\boldsymbol{\gamma},\boldsymbol{\beta})=\int_B K(\boldsymbol{\gamma},\boldsymbol{\beta};\boldsymbol{\gamma}',d\boldsymbol{\beta}') \tag{16}$$

for all $B\in\mathcal{B}(\mathbb{R}^{|\boldsymbol{\gamma}'|})$. In order to prove that *M* is $P(\boldsymbol{\gamma},\boldsymbol{\beta}|\mathbf{y})$-irreducible, it suffices to establish a μ-irreducibility of *M* for some σ-finite measure μ defined on the measurable space $([p]\times\mathbb{R}^p,2^{[p]}\otimes\mathcal{B}(\mathbb{R}^p))$, where $2^{[p]}$ denotes the power set of $[p]$ and $[p]=\{1,\ldots,p\}$, see [33]. Taking $\mu(\boldsymbol{\gamma},\boldsymbol{\beta})=p^{-1}\mathbb{I}_{\{1\}}(|\boldsymbol{\gamma}|)N(0,1)$, we want to show that, for any $\boldsymbol{\gamma}\in 2^{[p]}$ and $\boldsymbol{\beta}\in\mathbb{R}^{|\boldsymbol{\gamma}|}$, there is a non-vanishing probability for the state $(\boldsymbol{\gamma},\boldsymbol{\beta})$ to commute $(\{1,0,0,\ldots\},B)$ for every $\mu(\{1,0,0,\ldots\}\times B)>0$. Considering the one-step transition kernel (16) for the event that a death occurs, we have by construction

$$K(\boldsymbol{\gamma},\boldsymbol{\beta};\boldsymbol{\gamma}',d\boldsymbol{\beta}')=\frac{1}{3k}\min\{A,1\}\mathbb{I}_{S_{\boldsymbol{\gamma}',\boldsymbol{\beta}}}(\boldsymbol{\beta}')d\boldsymbol{\beta}', \tag{17}$$

where *A* is the first term in the death probability ratio in (11) and

$$S_{\boldsymbol{\gamma}',\boldsymbol{\beta}}=\{\boldsymbol{\beta}'\in\mathbb{R}^{k-1}|\exists j \text{ s.t. } \boldsymbol{\gamma}=\{j\}\cup\operatorname{supp}\boldsymbol{\beta}'\}.$$

Note that $\|\boldsymbol{\beta}\|_1=\|\boldsymbol{\beta}'\|_1+|u|\geq\|\boldsymbol{\beta}'\|_1$. Let $C_k=(\|\mathbf{x}^j\|_2^2u^2+2\langle\mathbf{y}-X\boldsymbol{\beta},\mathbf{x}^j\rangle u/\|\mathbf{y}-X\boldsymbol{\beta}\|_2^2)<\infty$ and $\alpha_k=(1+C_k)^{-1}$. The Cauchy–Schwartz inequality implies that $\alpha_k>0$. We have $A\geq C_k'\alpha_k>0$, where $C_k'=(p-(k-1)/(k-1)^2)(p(\boldsymbol{\gamma}'\to\boldsymbol{\gamma})/p(\boldsymbol{\gamma}\to\boldsymbol{\gamma}'))N(u;0,\varepsilon^2)>0$. So we deduce that

$$K(\boldsymbol{\gamma},\boldsymbol{\beta};\boldsymbol{\gamma}',d\boldsymbol{\beta}')\geq\frac{C_k'\alpha_k\mathbb{I}_{S_{\boldsymbol{\gamma}',\boldsymbol{\beta}}}(\boldsymbol{\beta}')\,d\boldsymbol{\beta}'}{3p}.$$

Iterating this process $k-1$ times, we can obtain

$$\begin{aligned}P(\{1,0,0,\ldots\}\times B|\boldsymbol{\gamma},\boldsymbol{\beta})&\geq\int_B\prod_{i=2}^{k}K(\boldsymbol{\gamma}_i,\boldsymbol{\beta}_i;\boldsymbol{\gamma}_{i-1},d\boldsymbol{\beta}_{i-1})\,d\boldsymbol{\beta}_1\\&\geq(3p)^{-(k-1)}\mu(\{1,0,0,\ldots\}\times B)\prod_{i=2}^{k}(C_i'\alpha_i)>0.\end{aligned}$$

The last step may be complemented by a standard MH-step (7) to update the coefficient with the same dimension. This shows that we can reach the state $(\{1,0,0,\ldots\},B)$ with a strictly positive probability. In summary, the above facts lead to the following convergence theorem.

Theorem 4.1. *Let* $(\boldsymbol{\gamma}^{(i)},\boldsymbol{\beta}^{(i)})$ *be the Markov chain with transition kernel given by the proposed RJ-MCMC algorithm in Algorithm*1. *This Markov chain converges to the posterior probability distribution* $\pi(\boldsymbol{\gamma},\boldsymbol{\beta}|\mathbf{y})$ *in* (6), *regardless of the initialization of the algorithm, i.e.*

$$\|\pi^{(i)}(\boldsymbol{\gamma},\boldsymbol{\beta})-\pi(\boldsymbol{\gamma},\boldsymbol{\beta}|\mathbf{y})\|_{TV}\to 0, \tag{18}$$

where $\pi^{(i)}(\boldsymbol{\gamma},\boldsymbol{\beta})$ *means the empirical distribution of* $(\boldsymbol{\gamma}^{(i)},\boldsymbol{\beta}^{(i)})$ *and* $\|\cdot\|_{TV}$ *means the total variation norm on bounded signed measures*,

$$\|\pi\|_{TV}=\sup_{B\in\mathcal{B}}\pi(B)-\inf_{B\in\mathcal{B}}\pi(B). \tag{19}$$

Remark 3. In addition to the birth-and-death proposal used in Algorithm 1, there is another main proposal studied in the literature, namely the *split-and-merge* strategy. It is worth noting that the convergence of the split-and-merge trans-dimensional strategy can also be established in a similar way. Theoretically, both proposals guarantee that the correspondingly designed algorithms converge to the right target distribution. The two proposals may result in different empirical convergence rates which are problem-dependent. As shown earlier, the adopted birth-and-death proposal yields satisfactory estimation performance in our simulations.

5. A diabetes data example

This is a benchmark data set used in [8]. It contains $n=442$ measurements from diabetes patients. Each measurement has ten baseline predictors: age, sex, body mass index (BMI), average blood pressure (BP), and six blood serum measurements (S1–S6). The response variable is a quantity that measures progression of the diabetes one year after baseline. The response is centered and the predictors are normalized to have zero means and unit variances, before applying any model selection methods.

Table 4
Estimated coefficients for the Lasso [32], Lar [8], the Bayesian Lasso based on the Gibbs sampler (GS) [27], BG [12], BG-MCMC, and proposed Bayesian Lasso for diabetes data.

Predictor	Lasso	Lar	GS	BG	BG-MCMC	Proposed BLasso
Age	−0.081	−0.325	−0.337	0	0	0
Sex	−10.920	−11.228	−11.043	−10.646	−10.923	−10.371
BMI	25.013	24.854	24.877	24.905	25.197	24.861
BP	15.074	15.303	15.183	15.380	15.633	14.655
S1	−15.803	−29.293	−20.997	−35.624	−29.872	−7.022
S2	5.196	15.970	9.551	25.314	20.377	0
S3	−4.498	1.232	−2.550	0	−2.778	−7.785
S4	5.839	7.434	6.174	0	0	3.168
S5	27.645	32.648	29.568	37.798	35.675	24.767
S6	3.101	3.174	3.202	0	0	2.227

We set the random walk step size of the proposed RJ-MCMC to 7 in order to control the acceptance ratio of proposed models to be around 30%. In our experiment, the acceptance ratio is 30.54%. We empirically observe that a smaller step size would cause the algorithm to explore the model space more slowly; while a larger one would have a higher rejection rate.

It is clear from Table 4 that models under comparison provide results with certain similarities, except for the predictors S1, S2, and perhaps S3. S1 selected by the proposed RJ-MCMC-based Bayesian Lasso has a negative coefficient with smaller magnitude than others. For S2, the estimated coefficients are less consistent across different models. For instance, Lasso and Gibbs sampler have positive coefficients with smaller magnitudes than those of the Lar and BG. For the proposed Bayesian Lasso, this predictor is essentially interpreted as insignificant. Therefore, it is not clear whether or not S1–S3 covariates should be selected from a solely computational point of view, and further medical or physical interpretation is needed to justify the choice of different models in a real-world problem.

6. A real fMRI application

6.1. Application description

In this section we demonstrate an application to real fMRI data derived from subjects with Parkinson's Disease (PD). The problem of interest here is to employ the sparse linear regression modeling for learning brain functional connectivity using fMRI data.

The fMRI data we have are from 10 normal people and eight PD patients. During the fMRI experiment, subjects continually squeezed a bulb in their right hand to control an inflatable ring so that the ring moved through an undulating tunnel without touching the sides. A trial of the task was 5 min. Normal subjects performed only one trial; the PD subjects performed the same task twice, once before medication, the other after the medication. fMRI data were collected with a Philips Achieva 3.0 T scanner with a TR interval of 2 s. Hence, we collected 150 data points for each subject. After motion correction, the fMRI time courses of the voxels within each ROI were averaged to represent the summary activity of each ROI. The averaged time courses were then linearly detrended and normalized to unit variance. In this study, based on previous neuroscience knowledge, 18 brain regions were selected as the ROIs based on each person's individual anatomy.

The model that we assume here is a linear regression one which incorporates both spatial and temporal effects of brain connectivities. To do so, we combine the structural equation modeling (SEM) [23] and multivariate autoregressive model (mAR) [15]. Specially, let $\mathbf{y}(t)=(y_1(t),\ldots,y_p(t))^T$ be a $p\times 1$ dimensional vector, which contains the intensity measurements for the p brain ROIs at time t, for $t=1,\ldots,T$. For mAR component of the unified model, we consider only up to an mth order process. Hence, by combining the SEM part, the joint SEM and mAR model can be expressed as

$$\mathbf{y}(t)=\underbrace{\boldsymbol{A}\mathbf{y}(t)}_{\text{SEM}}+\underbrace{\sum_{j=1}^{m}\boldsymbol{\Phi}(j)\mathbf{y}(t-j)}_{\text{mAR}}+\underbrace{\mathbf{e}(t)}_{\text{noise}}, \tag{20}$$

where $\boldsymbol{A}$ and $\boldsymbol{\Phi}(j)$ are $p\times p$ coefficient matrices to be estimated, for $j=1,\ldots,m$; $\mathbf{e}$ is a noise vector which is assumed to be Gaussian with zero mean and constant variance σ^2. $\boldsymbol{A}$ represents the spatial connection strengths between ROIs, while $\boldsymbol{\Phi}(j)'$ s are the time lag effect strengths. We are aiming at applying the proposed fully Bayesian method to simultaneously select the model order and estimate the functional connectivities between brain regions represented by the coefficients.

6.2. Results

We primarily want to check the consistency of the estimates from the aforementioned algorithms. First, we plot and examine the correlations between the estimated coefficients of these algorithms. The result is shown in Fig. 2. We can see that the proposed Bayesian Lasso, Lar, and Lasso estimates are highly correlated. In particular, the Lar and Lasso estimates have a higher similarity (with the correlation coefficient being 0.967) than being compared with the proposed Bayesian Lasso (with the correlation coefficient being 0.873 and 0.870 for the Lar and Lasso estimates, respectively).

We further investigate estimation stability. We take one subject's fMRI time-series data and split the 150 time data points into two subsets of size 100, with the middle 50 data points overlapped. We then learn two models from these two subsets and examine the correlation

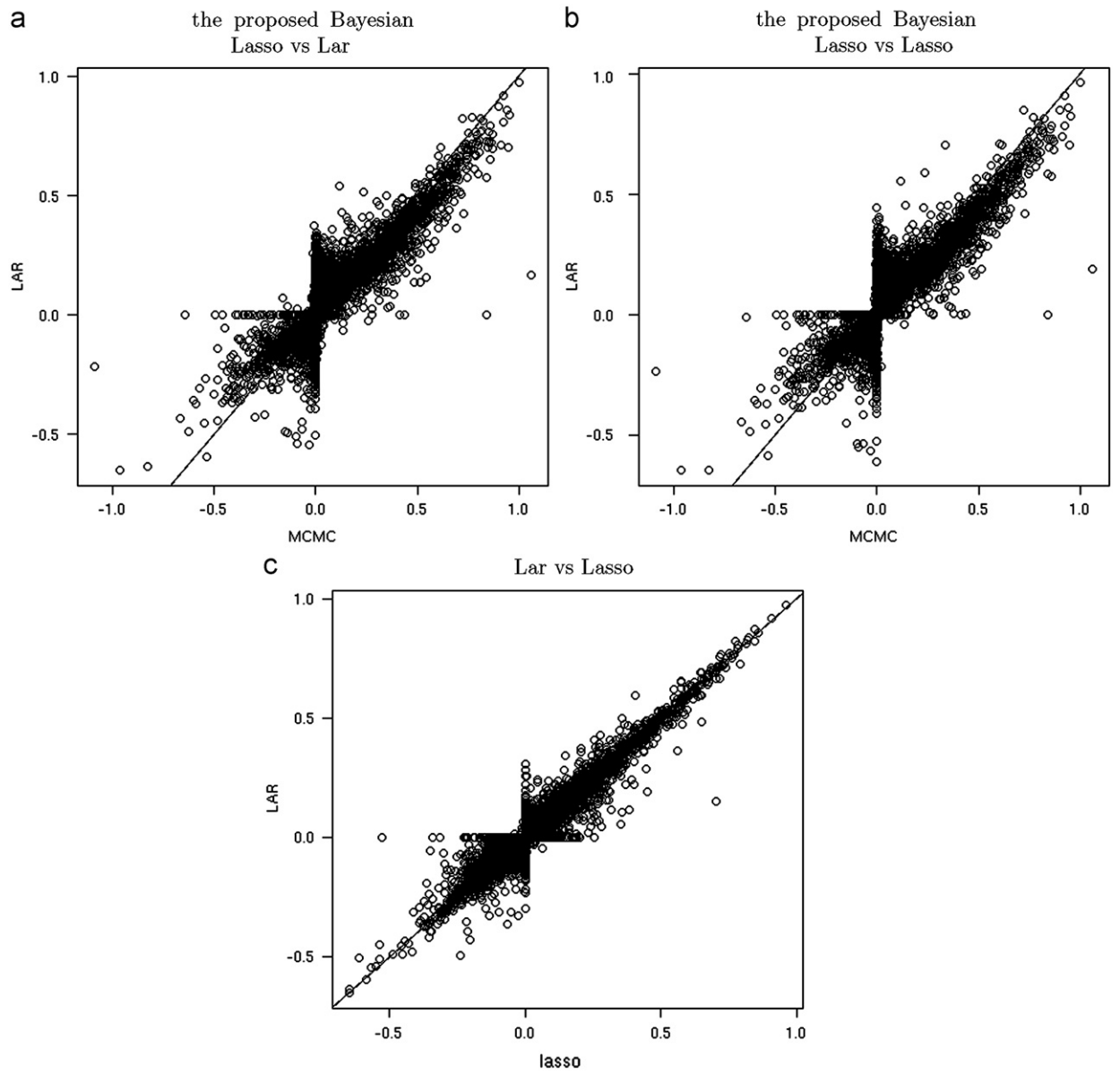


Fig. 2. The correlation between the estimated coefficients when using different algorithms. (a) The proposed Bayesian Lasso vs Lar; (b) the proposed Bayesian Lasso vs Lasso; (c) Lar vs Lasso.

between the estimated coefficients from subsets. The MLE approach for the full (non-sparse) model is also included for comparison. It is noted that the proposed Bayesian Lasso reveals greater estimation stability between models derived from subsets of the fMRI data compared to the other three methods, since it yields the highest similarity between the model coefficients derived from the two subsets (represented by a correlation coefficient being 0.702, see Table 5). The Lar and Lasso estimates yield a lower correlation (with the correlation coefficient being 0.599 and 0.587, respectively). The MLE approach provides the lowest consistency (with the correlation coefficient being 0.293), which is not a surprising fact since the variability of the MLE estimate is usually larger than the estimates via sparse regression. Moreover, limiting our attention only to the predictor coefficients which are estimated as non-zero from both subsets, we note that the correlations between the estimates derived from two subsets are even higher for the proposed Bayesian Lasso, Lar, and Lasso approaches.

Table 5
Correlations between the coefficient estimates on two fMRI data subsets when using MLE, Lar [8], Lasso [32] and the proposed BLasso.

Method	MLE	Lar	Lasso	Proposed BLasso
Correlation	0.293	0.599	0.587	0.702

7. Discussion and conclusion

In this paper, we proposed a hierarchical, fully Bayesian version of the Lasso model for inferring sparse linear regression from high-dimensional data sets. Since the joint posterior distribution of the parameters involves both discrete and continuous parameters, we developed a reversible-jump MCMC algorithm to compute the unbiased minimum variance estimates. Simulations demonstrated that the proposed Bayesian Lasso estimate yields lower estimation errors when compared with popular Lasso-type estimates and a Gibbs sampler-based Bayesian estimate. One intuitive explanation of this observation is that model averaging by the fully Bayesian approach provides better stability than selecting only a single *best* model. The simulations further demonstrated that the proposed Bayesian Lasso is robust to correlated predictors, even though the hypothesis of independent priors for predictors is assumed in the model design (3). We proved the convergence of the proposed RJ-MCMC algorithm for the Bayesian Lasso. Further, we extended the proposed fully Bayesian framework to the Binomial–Gaussian model, and simulations showed that the proposed stochastic search could substantially improve the performance of the original BG model-based estimate in [12].

One important direction for future work is to improve the sampling strategy. Currently, we use the Gaussian RW proposal with a fixed variance parameter. However, as shown in [16], the adaptive RJ-MCMC sampler usually facilitates mixing speed, and thus a data-driven adaptive sampler is of particular interest. We also observe that different step sizes can lead to different models and affect the empirical convergence.

The proposed RJ-MCMC Bayesian Lasso approach also has limitations. One is its higher computational cost compared with Lasso-type estimates, a limitation common to MCMC-based Bayesian approaches. This limitation prevents using the proposed method from online high-dimensional estimation problems. However, for offline/batch estimation problems (e.g. fMRI modeling), the proposed method can usually provide practical accurate estimates with affordable computational complexity. Table 6 reports the required CPU times by different methods in the case of $p=15$ and uncorrelated design. We observed similar empirical complexity results for other settings in our simulations. Basically, among the discussed MCMC-based Bayesian approaches, we note that the Gibbs sampler-based Bayesian Lasso requires the highest computational cost, followed by BG-MCMC, while the RJ-MCMC Bayesian Lasso requires the least computational time. In summary, the computational costs of different estimates can be ordered as

$$\text{Lar} \prec \text{Lasso} \prec \text{BG} \prec \text{RJ-MCMC Bayesian Lasso} \prec \text{BG-MCMC} \prec \text{Gibbs sampler based Bayesian Lasso}.$$

Table 6
The required CPU times for Lar [8], Lasso [32], the Bayesian Lasso-based on the Gibbs sampler (GS) [27], BG [12], BG-MCMC, and proposed Bayesian Lasso, in the case of $p=15$ and uncorrelated design. CPU times are normalized w.r.t. the Lar running time. The last three simulation based methods run 20,000 iterations.

Method	Lar	Lasso	BG	BG-MCMC	RJ-MCMC	GS
CPU time	1.000	1.296	25.250	346.065	98.749	2848.318

Acknowledgments

We thank two anonymous reviewers for their constructive suggestion that significantly improved an earlier version of the paper. This work was supported by a Pacific Alzheimer's Research Foundation (PARF) Centre Grant Award.

Appendix A. Derivation of the joint posterior distribution in (5)

Denote the likelihood of the model by $L(\mathbf{y}|X,\cdot)$. We can factorize the joint posterior distribution according to the conditional independence relationships encoded in the hierarchical model as

$$\begin{aligned}
&p(\boldsymbol{\beta},\tau,\sigma^2,\boldsymbol{\gamma},\lambda|\mathbf{y},X)\\
&\propto p(\boldsymbol{\beta},\tau,\sigma^2,\boldsymbol{\gamma},\lambda)\times L(\mathbf{y}|X,\boldsymbol{\beta},\tau,\sigma^2,\boldsymbol{\gamma},\lambda)\\
&\quad\propto \frac{1}{\binom{p}{k}}p(\lambda)p(\tau,\sigma^2)p(k|\lambda)\prod_{j\in\boldsymbol{\gamma}}p(\beta_j|\boldsymbol{\gamma},\tau)\\
&\times(\sigma^2)^{-n/2}\exp\left(-\frac{\|\mathbf{y}-X\boldsymbol{\beta}\|_2^2}{2\sigma^2}\right)\\
&\quad\propto \frac{1}{\binom{p}{k}}(\lambda\tau\sigma)^{-1}\frac{e^{-\lambda}\lambda^k}{k!}\tau^{-k}\exp\left(-\frac{\sum_{j\in\boldsymbol{\gamma}}|\beta_j|}{\tau}\right)\\
&\times\sigma^{-n}\exp\left(-\frac{\|\mathbf{y}-X\boldsymbol{\beta}\|_2^2}{2\sigma^2}\right),
\end{aligned}$$

which gives (5).

Appendix B. MCMC algorithm for the Binomial–Gaussian model

We apply the proposed fully Bayesian framework to the Binomial–Gaussian model proposed in [12]. More specifically, let $\boldsymbol{\gamma}$ be a p-length binary vector representing an active set. The active set is assumed to be binomially distributed

$$p(\boldsymbol{\gamma}|w)=w^{q_{\boldsymbol{\gamma}}}(1-w)^{p-q_{\boldsymbol{\gamma}}}, \tag{B.1}$$

where w is the probability of taking value one and $q_{\boldsymbol{\gamma}}$ means the number of ones in $\boldsymbol{\gamma}$. Conditioning on $\boldsymbol{\gamma}$, Zellner's g-prior is used for the coefficient:

$$p(\boldsymbol{\beta}_{\boldsymbol{\gamma}}|\boldsymbol{\gamma},g)=N(\mathbf{0},g\sigma^2(X_{\boldsymbol{\gamma}}^TX_{\boldsymbol{\gamma}})^{-1}). \tag{B.2}$$

Then, assigning a conjugate beta prior on w and Jeffrey's non-informative priors on σ^2 and g, we have

$$p(w)=\text{Beta}(a,b),$$

$$p(\sigma^2,g)\propto(\sigma^2g)^{-1}.$$

Therefore, we can analytically integrate out $\boldsymbol{\beta}_{\boldsymbol{\gamma}}$ and obtain

$$\begin{aligned}
&p(\mathbf{y},g,\sigma^2,w,\boldsymbol{\gamma})\propto p(w)p(g)p(\sigma^2)p(\boldsymbol{\gamma}|w)p(\mathbf{y}|g,\sigma^2,\boldsymbol{\gamma})\\
&\propto w^{a-1}(1-w)^{b-1}(\sigma^2)^{-1}g^{-1}w^{q_{\boldsymbol{\gamma}}}(1-w)^{p-q_{\boldsymbol{\gamma}}}
\end{aligned}$$

$$\times(2\pi\sigma^2)^{-n/2}(1+g)^{-q_\gamma/2}\exp\left(-\frac{\mathbf{y}^T\mathbf{y}}{2\sigma^2}+\frac{g}{2\sigma^2(1+g)}R_\gamma\right), \tag{B.3}$$

where $R_\gamma=\hat{\boldsymbol{\beta}}_\gamma^T X_\gamma^T X_\gamma\hat{\boldsymbol{\beta}}_\gamma$ and $\hat{\boldsymbol{\beta}}_\gamma$ is least squares estimate of model γ. By further integrating out σ^2 and w, we obtain

$$p(\mathbf{y},g,\gamma)\propto B(a+q_\gamma,b+p-q_\gamma)g^{-1}(1+g)^{-q_\gamma/2}\left(-\frac{\mathbf{y}^T\mathbf{y}}{2}+\frac{g}{2(1+g)}R_\gamma\right)^{-n/2} \tag{B.4}$$

where $B(a+q_\gamma,b+p-q_\gamma)=\Gamma(a+q_\gamma)\Gamma(b+p-q_\gamma)/\Gamma(a+b-p)$.

Now, it follows that the full conditional probabilities

$$p(\gamma|\mathbf{y},g)\propto B(a+q_\gamma,b+p-q_\gamma)(1+g)^{-q_\gamma/2}\left(-\frac{\mathbf{y}^T\mathbf{y}}{2}+\frac{g}{2(1+g)}R_\gamma\right)^{-n/2}, \tag{B.5}$$

$$p(g|\mathbf{y},\gamma)\propto g^{-1}(1+g)^{-q_\gamma/2}\left(-\frac{\mathbf{y}^T\mathbf{y}}{2}+\frac{g}{2(1+g)}R_\gamma\right)^{-n/2}\mathbb{I}(g\geq 0). \tag{B.6}$$

Thus the updating probability from $\gamma\to\gamma'$ is the MH ratio

$$\min\left\{\frac{p(\gamma'|\mathbf{y},g)}{p(\gamma|\mathbf{y},g)}\times\frac{p(\gamma'\to\gamma)}{p(\gamma\to\gamma')},1\right\}. \tag{B.7}$$

We note that the above BG-MCMC algorithm can be interpreted as an RJ-MCMC approach with the birth-and-death proposal. Here since the Jacobian for the birth-and-death proposal is 1 for model jumping, there is no explicit Jacobian term reflected in the generalized MH ratio and thus it is coincides with the ordinary MH ratio with γ being considered as a regular parameter.

Regarding the parameter g, we employ a symmetric random walk sampler for updating. More specifically, we let $g'=g+u$ where $u\sim N(0,\varepsilon^2)$ and accept with probability $\min\{p(g'|\mathbf{y},\gamma)/p(g|\mathbf{y},\gamma),1\}$.

The MCMC algorithm for the BG model is summarized in Algorithm 2.

Algorithm 2. BG-MCMC algorithm

Input: The number of iterations T. Random walk step size ε.
Data: X and $\mathbf{y}$.
Output: $\{\{\gamma^{(t)},g^{(t)}\}|t\in\{0,\ldots,T\}\}$.

```
begin
    Initialization : choose{γ^(0), g^(0)} and t = 1.
    repeat
        if k^(t−1) = 1 then
            k^(t) ← k^(t−1) + 1.
        else if k^(t−1) = p then
            k^(t) ← k^(t−1) − 1.
        else
            k^(t) ← k^(t−1) + U({−1, 1}).
        end
        if k^(t) = k^(t−1) + 1 then
            Propose a γ' such that |γ'| = |γ| + 1 (birth move);
            Accept γ' with the probability given by MH ratio.
        else
            Propose a γ' such that |γ'| = |γ| − 1 (death move);
            Accept γ' with the probability given by MH ratio.
        end
        Sample g with the Gaussian random walk with step size ε.
        t ← t + 1.
    until t = T.
end
```

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